

Radiotherapy for thyroid cancer

External beam radiotherapy is used to treat some types of thyroid cancer. Radiotherapy uses high energy rays called radiation to destroy cancer cells.

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External beam radiotherapy for thyroid cancer

[External beam radiotherapy](#) uses high energy rays called radiation to treat cancer. It destroys cancer cells in the area where the radiotherapy is given.

This treatment is used in different ways depending on the [type of thyroid cancer](#) you have.

Follicular and papillary thyroid cancer

Radiotherapy may be used for [follicular and papillary thyroid cancer](#) if it is not possible to remove all of the tumour with [surgery](#). It may also be used if the cancer has come back in the neck area or if the cancer has spread to other parts of the body, for example to the bones.

Medullary thyroid cancer (MTC)

People with [medullary thyroid cancer \(MTC\)](#) may sometimes be given radiotherapy after [surgery](#) to try to reduce the risk of the cancer coming back. You may also have radiotherapy to improve symptoms or if the cancer has spread to other parts of the body.

Anaplastic thyroid cancer

Radiotherapy may be used to treat [anaplastic thyroid cancer](#). It can be given on its own or with chemotherapy ([chemoradiation](#)).

Radiotherapy may be used to shrink the tumour before surgery. It may also be given after surgery to reduce the risk of the cancer coming back. It can also help to control any symptoms.

External beam radiotherapy does not make you radioactive. It is safe for you to be with other people, including children, throughout your treatment.

Planning your treatment

Before you start your treatment, it needs to be carefully planned. You will have a hospital appointment to plan your treatment.

Planning makes sure that the radiotherapy is aimed precisely at the cancer. This means it causes the least possible damage to the surrounding healthy tissue.

If your neck area is being treated, you will need to have a [mould or mask](#) made before your treatment is planned. This is to keep your head still while you have treatment. You will also have a [CT scan](#) to help with planning.

Having external beam radiotherapy

Treatment sessions

You normally have external radiotherapy as a series of short, daily treatments in a hospital radiotherapy department. A radiotherapy machine aims high energy rays at the area of the body being treated.

You usually have radiotherapy as an outpatient. How many treatment sessions you have depends on the aim of your treatment. You then have a rest with no treatment at the weekend.

The person who operates the machine is called a radiographer. They give you information and support during your treatment.

Positioning you for treatment

At the beginning of each session, the radiographers will make sure you are in the correct position and carefully fit [your mask](#) if you have one. They will tell you how long your treatment will take. When everything is ready, they leave the room and the treatment starts.

The treatment itself is not painful. It does not make you radioactive. You will not be a risk to anyone during the course of your treatment.

The radiographers can see and hear you from outside the room. There is usually an intercom, so you can talk to them if you need to during your treatment.

Side effects of external beam radiotherapy

Radiotherapy to the neck can cause [side effects](#). This will depend on how much treatment you are having, and the area being treated.

Most side effects have usually improved about 6 weeks after radiotherapy has ended. Your cancer doctor, specialist nurse or radiographer will discuss possible side effects with you before you start your treatment.

Always tell your radiotherapy team about any side effects. There are usually things they can do to help improve them. They will give you advice on what you can do to manage side effects. For example, they can give you advice on how to:

- care for the skin in the treated area
- take care of your mouth
- check for any signs of infection, such as oral thrush.

The side effects will depend on the part of the body being treated and how much treatment you are having. If you have radiotherapy to improve symptoms the side effects are often mild. You may need only a few treatments.

If you are having radiotherapy to the neck, side effects can include the following:

- [Tiredness](#) – this can continue for some weeks or longer after treatment finishes. Balance rest with some physical activity. Even going for short walks will help you to feel less tired.
- [Discomfort or pain when swallowing](#).
- [Skin changes](#) in the area being treated – the skin in the treated area may become sore or itchy. If you have white skin, the skin may redden. If you have black or brown skin, the skin may become darker than the surrounding skin.
- [A dry throat, and sometimes mouth](#) – sip water regularly and avoid eating dry foods, ask your team about artificial saliva products they can prescribe for you.
- Thick, sticky saliva – using a sodium bicarbonate mouthwash may help clear thick saliva. Your radiotherapy team for advice about the type of mouthwash that is best for you, and how to prepare it.
- [Taste changes](#) – this can take some weeks or longer after treatment finishes to improve. Try different foods to find out what tastes best to you.

Your doctor can give you information to help you cope with any side effects.

Radiotherapy to the neck may also cause [late effects](#). These are side effects that do not go away, or that develop months or years after treatment.

Your doctor can tell you about any possible late effects and what you can do to help reduce the risk of these.

Getting support

Macmillan is here to support you. If you would like to talk, you can do the following:

- Call the Macmillan Support Line for free on [0808 808 00 00](tel:0808 808 00 00).
- [Chat to our specialists online](#).

Visit our [radiotherapy and side effects forum](#) on the Online Community to connect with people affected by radiotherapy and share your experience.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our thyroid cancer information. If you would like more information about the sources we use, please contact us at

informationproductionteam@macmillan.org.uk

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help